DESCRIPTION OF COURSEWORK

Course Code	EEE302
Course Name	Electromagnetic Field Theory and Waves
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2024/09
Assessment Title	Lab report: Full Two-port Microwave Calibration Technique

A. Introduction/ Situation/ Background Information

S-parameters (scattering parameters) are used to describe the properties of a microwave component and transmission line. The FieldFox Network Analyzer is modern measurement tool that has the capability to measure all the four S-parameters with a single connection.

S11 and S22 reflection measurements are used to measure the amount of reflections off the corresponding DUT port. Low reflections means there is a good impedance match between the source and DUT. S21 and S12 transmission measurements are used to measure the loss or gain through a DUT over a specified frequency range. Both ends of the DUT must be connected to the FieldFox. The FieldFox signal source is transmitted out one of the test port connectors, through the DUT, and into the other test port connector.

Microwave Calibration Standards

Calibration removes the systematic errors that are associated with measurements. There are well-defined and understood systematic errors that are measured and calculated during the calibration process. These errors are caused by leakage signals inside the FieldFox, by the frequency response of the FieldFox receivers, and by reflections inside the FieldFox that interact with the DUT. After calibration, these errors are removed from subsequent measurement

Calibration Standards (THRU,OPEN, SHORT, and LOAD) are standards that are used during calibration. When an RF signal are incident on these components, the signals are reflected in a predictable manner. These components can also be used to terminate a DUT port during some measurements. SHORT and OPEN standards both cause 100% of an RF signal to be reflected. The

difference between these two standards is what happens to the phase of the reflected signal, which is beyond the scope of this discussion. Although an OPEN standard is a precision component, simply leaving nothing connected at the end of a cable can be a reasonable substitute for an OPEN. A LOAD standard absorbs almost ALL of the incident signal and very little signal is reflected back to the source. A THRU standard is used during some calibration steps to connect PORT 1 to PORT 2 in place of the DUT. A Flush THRU connection can be made when cables that connect with the DUT can mate with each other.. Otherwise, any reasonably short cable can be used as a THRU standard. Calibration Reference Plane is the point (or points) at which the DUT and cal standards are connected during a calibration. This can be at the FieldFox test port connectors, or at the end of jumper cables or adapters

Assignments for this coursework:

Full Two-port Microwave Calibration Technique (lab report)

In this experiment, the students shal perform a full two port calibration on a N9913A FieldFox Network Analyzer. This assignment is designed for students to apply the basic concepts of impedance matching problem between the transmission lines and open, short ,load and thru of different impedances. The Fieldfox has a builtt-in software to determine the S-parameter values from an error correction algorithm using the measured impedance values of the open, short, load and thru standards.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Measure/determine the characteristic of transmission lines using instruments and software or other specified tools

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

The report must be submitted (softcopy/ hardcopy) before 14 July 2024. The reports must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Full Two-port Microwave Calibration Technique	20%
	TOTAL	

F. Tasks

1. Perform measurements of the open,short, load at the end of the cables at port 1.
2. Repeat the measurement at port 2.
3. Measure the thru standard by connecting the cables from port 1 and port 2 together.
4. Verify the measurement by measuring the magnitude and phase of S_{11}
5. Save all measurement data either in ASCII or S2p format
6. Plot the graphs of magnitude and phase for S_{21} vs. frequency and S_{11} vs. frequency
7. Apply the theory electromagnetic wave propagation to analyse all the measurement results obtained using the Fieldfox analyzer

COMPLEX ENGINEERING PROBLEM

Description of Complex Engineering Problem	
Project Theme: Full Two-port Microwave Calibration Technique **Assessment: Presentation/Report/Demonstration**	
Attributes	**Remarks**
WP1 Depth of knowledge required Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach.	Requires students to analyse Fieldfox microwave measurement results using the theory of electromagnetic waves **(WK6)**
WP2 Range of conflicting requirements Involve wide-ranging or conflicting technical, engineering and other issues.	Students will need to distinguish between 1. Open,short, load and thru standards 2. S_{11}, S_{21}, S_{12} and S_{22}
WP3 Depth of analysis required Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	Students need to use theory of electromagnetic waves and transmission line theory to determine the relationship 1.between the S-parameter, incident electric field(voltage), reflected electric field(votage), transmitted electric field(voltage) 2.the expected values of the reflection coefficient of open, short, load and thru standards. 3, the expected value of the thru standard
WP7 Interdependence Are high level problems including many component parts or sub-problems	Calibartion of network analyzer is a high level problem which have different components: the theory of wave propagation, transmission line theory, the network analyzer skills required for the measurement procedure, results and analysis.

No.	EAC PO Attribute	Description	Bloom's Taxonomy Domains	WK	WP/ EA
PLO5 / CLO3	**Modern Tool Usage**	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations	Psychomotor	WK6	WP

APPENDIX 1

MARKING RUBRICS

	Excellent	Good	Adequate	Poor	Lacking
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.
methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized

Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.
Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or real world connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.

APPENDIX 1

MARKING RUBRICS

Component Title	ASSIGNMENT				Percentage (%)	20	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Abstract						10%	2
Introduction & Theory						10%	2
Methodology/ Experimental						40%	8
Results, Disscussion Conclusion, Formatting References						40%	8
					TOTAL	100%	20

Criteria	Score and Descriptors				
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**
Abstract	Clear, concise, and thorough summary including context, important results, and conclusions.	Refers to most of the major points, but some minor details are missing or not clearly explained	Misses one of more major parts of the results, context, or conclusions	Missing several majors aspects and merely repeats information from the introduction	None or unrelated
Introduction	A cohesive, well-written summary (including all relevant chemistry) of the background material pertinent to the experiment with appropriate references. Places the purpose of the experiment in context.	Is nearly complete but does not provide context for minor points. Contains relevant information but fails to provide background for one aspect of the experiment, or certain information is not cohesive.	Certain major introductory points are missing (ex: background, theory, chemistry, context, etc.) or explanations are unclear and confusing. References are used properly.	Very little background information is provided and/or information is incorrect. No references are provided.	None or unrelated
Experimental	Contains details on how the experiment was performed and the procedures followed. Written in the correct tense and omits information that can be assumed by peers (trained chemists)	Narrative includes most important experimental details but is missing one or more relevant pieces of information.	Missing several experimental details or some incorrect statements.	Several important experimental details are missing. Narrative is incorrect, illogical, or copied directly from the lab manual. Written in the incorrect tense.	None or unrelated
Results (Presentation of results, figures and tables)	All figures, graphs, and tables are numbered with appropriate captions. All tables, figures, etc. are explicitly mentioned in the text. Relevant experimental data are presented which are used in the discussion.	All figures, graphs, and tables are correctly drawn, but some have minor problems that could be still be improved. All data and associated figures, etc. are mentioned in the text. Most relevant data present.	Most figures, graphs, and tables are included, but some important or required features are missing. Certain data reported are not mentioned in the text or are missing. Captions are not descriptive or incomplete.	Figures, graphs, and tables are poorly constructed; have missing titles, captions or numbers. Certain data reported are not mentioned in the text. Important data missing.	None or unrelated